

SOFTWARE PROJECT MANAGEMENT

Unit 1

Project & Project Management (VERY IMPORTANT)

- Definition of Project
- Definition of Project Management
- Key characteristics of a project
- Objectives of project management

Project Management Framework

- Meaning of project management framework
- Major components (phases / processes)
- How it helps in managing projects

Role of Project Manager

- Definition of project manager
- Roles and responsibilities
- Skills required (technical, managerial, leadership)

Project Life Cycle

- Definition
- Phases (Initiation, Planning, Execution, Monitoring, Closure)
- Importance of each phase
- Diagram might be better

Stakeholder Management

- Definition of stakeholder
- Types of stakeholders
- Importance of stakeholder management

Expected Questions from Unit–1 (Very Likely)

- Define project and project management
- Explain role of project manager
- Explain project life cycle
- Describe project management framework
- Short note on stakeholder management / leadership styles

Unit 2

Waterfall Model (TOP PRIORITY)

- Definition
- Phases (in order)
- Advantages
- Disadvantages
- When to use
- Diagram strongly recommended

Prototype Model (Evolutionary Model)

- Meaning of prototyping
- Working / steps
- Advantages
- Disadvantages
- Applications

Spiral Model (VERY IMPORTANT)

- Definition
- Spiral cycles
- Risk analysis concept
- Advantages
- Disadvantages

SCRUM (EXTREMELY IMPORTANT)

- Introduction to Scrum
- Three pillars of empiricism
 - o Transparency
 - o Inspection
 - o Adaptation
- Scrum framework (basic flow)
- Artifacts (Product backlog, Sprint backlog, Increment)
- Role of Scrum Master

Agile Development Model (Conceptual)

Lean Model

DevOps Model

Waterfall vs Agile / Waterfall vs Spiral

MOST LIKELY QUESTIONS FROM UNIT-2

- Explain Waterfall model
- Explain Spiral model with risk analysis
- Explain Scrum framework / pillars of empiricism
- Compare Waterfall and Agile

Unit 3

Types of Requirements (HIGH PRIORITY)

- Functional requirements
- Non-functional requirements
- User requirements
- System requirements

Feasibility Study (EXTREMELY IMPORTANT)

- Meaning of feasibility study
- Types:
 - o Technical
 - o Economic
 - o Operational
 - o Legal
 - o Schedule

Requirement Elicitation Techniques (VERY IMPORTANT)

- Meaning of requirement elicitation
- Techniques (2–3 lines each):
 - o Interview
 - o Questionnaire
 - o Brainstorming
 - o FAST

SRS (Software Requirement Specification) (TOP SCORING)

- Definition of SRS
- Contents of SRS
- Characteristics of good SRS

DFD (Data Flow Diagram)

UML Overview & Purpose

Use Case Diagram

Class Diagram

MOST LIKELY QUESTIONS FROM UNIT–3

- Explain feasibility study
- Explain SRS and its contents
- Explain requirement elicitation techniques
- Draw and explain DFD
- Draw and explain Use Case diagram
- What is UML? Explain its purpose

Unit 4

Business Case (MEDIUM PRIORITY)

- Meaning of business case
- Purpose
- Contents (benefits, costs, risks)

Scope Management (HIGH PRIORITY)

- Meaning of scope
- Importance
- Scope planning and control (brief)

Function Point Analysis (FPA)

COCOMO I

COCOMO II

MOST LIKELY QUESTIONS FROM UNIT-4

- Numerical on Function Point
- Numerical on COCOMO model
- Explain scope management
- Explain COCOMO II models
- Short note on business case

Unit 5

Critical Path Method (CPM)

- Meaning of CPM
- Network diagram basics
- Terms:
 - o ES (Earliest Start)
 - o EF (Earliest Finish)
 - o LS (Latest Start)
 - o LF (Latest Finish)
 - o Float / Slack
- How to identify Critical Path

Work Breakdown Structure (WBS)

- Definition of WBS
- Steps to create WBS
- Advantages
- Simple example

Project Schedule & Schedule Control

- Meaning of project scheduling
- Schedule control techniques
- Why schedule control is needed

Resource Allocation

Procurement Management

Project scheduling tools

MOST LIKELY QUESTIONS FROM UNIT-5

- Solve a CPM numerical
- Explain WBS
- Effect of schedule change on cost
- Explain change management
- Short note on procurement management

Unit 6

Software & System Quality Management

- Meaning of software quality
- Meaning of quality management
- Importance of quality in software projects

Six Sigma

- Meaning of Six Sigma
- Objective (defect reduction)
- DMAIC phases (just names + one line)

Formal Technical Reviews

- Meaning
- Purpose
- Benefits

Risk Analysis & Assessment

- Risk probability
- Risk impact
- Risk exposure concept

Risk Strategies

- Risk avoidance
- Risk mitigation
- Risk transfer
- Risk acceptance

Risk Monitoring, Control, Response & Evaluation

- Meaning of monitoring
- Why control is needed
- Risk response planning
- Risk evaluation

MOST LIKELY QUESTIONS FROM UNIT-6

- Explain ISO 9001
- Explain CMM with levels
- Explain software risk management
- Risk analysis and strategies
- Short note on Six Sigma